Research: Cold Start Problem Taxonomy

Date: April 9, 2026 **Report:** `20260409.01_cold-start-network-effects` **Research Question:** What are the distinct types of cold start problems, and how do you diagnose which type you're facing?

Executive Summary

The "cold start problem" is not one problem — it is a family of related but distinct challenges that depend on the product type, network structure, and what kind of value is missing. Andrew Chen's framework from "The Cold Start Problem" (2021) is the most comprehensive treatment available. He doesn't create a rigid taxonomy of "four types," but the book and surrounding practitioner literature make it possible to construct one from first principles.

The key insight for student teams: **the solution depends entirely on the diagnosis.** A marketplace cold start requires finding supply. A content cold start requires seeding or enabling creators. A group/quorum cold start requires coordinated adoption. A network cold start requires density within a bounded community. An AI/algorithm cold start requires bootstrapping training data. Getting the diagnosis wrong means applying the wrong playbook.

Source Index

All sources used in this report, with URLs:

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Part 1: Andrew Chen's Core Framework — "The Cold Start Problem" (2021)

1.1 The Central Concept

Andrew Chen, General Partner at a16z and former growth leader at Uber, published "The Cold Start Problem" in December 2021. The book is based on interviews with founding teams from LinkedIn, Slack, Airbnb, Zoom, Tinder, Clubhouse, Reddit, YouTube, and others, plus Chen's own experience at Uber.

The central argument: most networked products die at birth because of **anti-network effects** — the destructive mirror image of network effects. Before a network reaches critical mass, it actively repels users. Open an empty social feed and you leave. Open Uber with no drivers and you never come back. The product fails not because of the product itself, but because the network is too sparse to deliver value.

Chen calls this the **Cold Start Problem**: "When a networked product launches, it faces a chicken-and-egg problem: people need to use it for it to be worth anything." (andrewchen.com)

The book's central insight: **network density beats network size.** A small, densely connected network is more valuable and more survivable than a large, sparse one. The goal isn't to get as many users as possible — it's to get the right users in the right configuration.

1.2 The Atomic Network

The atomic network is the central operational concept in the book. It is defined as **the smallest network that is stable and can grow on its own** — the minimum viable network that must be both dense enough and complete enough to deliver real value before growth begins.

Key characteristics:

- **Remarkably small in practice.** For Zoom, two people. For Slack, one company team. For Airbnb, hundreds of listings in a single city. For credit cards, an entire commercial district. The size varies enormously by product type.
- **Defined by stability, not size.** An atomic network is stable when the users in it keep coming back without constant external intervention — when the "perpetual motion machine" starts turning on its own.
- **Hyper-specific by definition.** Chen recommends targeting something like "the Q2 planning cycle in the Product team at Chase Bank" rather than "enterprise software buyers." The more specific the target, the higher the density. (lennysnewsletter.com/p/atomic-network)

The "copy and paste" scaling strategy: once one atomic network is stable, you build a second adjacent network using the same playbook. Each successive network is easier because interconnections strengthen. Uber went from one city to dozens to hundreds using this model. Facebook went from one campus to 2,200 campuses to the world. (lennysnewsletter.com/p/atomic-network)

Atomic network size examples by product:

Product	Atomic Network
Zoom	2 people wanting to video call
Slack	1 company's internal team
Airbnb	300 homes + 100 reviews in one city
Uber (downtown LA)	~15 concurrent drivers within range
Facebook	Harvard student body
Tinder	USC campus population (seeded by events)
Credit card (AmEx)	All merchants in one commercial district

1.3 Hard Side vs. Easy Side

Every networked product has two types of users with **asymmetric difficulty** and **asymmetric value creation.**

The Hard Side is the small percentage of users who do the majority of the work and create disproportionate value for everyone else. They are difficult and expensive to recruit. They have options. They are often professionals or semi-professionals with real opportunity costs. But without them, the product delivers zero value.

The Easy Side follows once the hard side is in place. They need the hard side to exist before they see any reason to show up.

Crucially: **solving the cold start problem means solving for the hard side first.** Every strategy, subsidy, and early-stage tactic should be aimed at the hard side, not the easy side. (chrismillas.com)

Hard side examples by product type:

Product	Hard Side	Easy Side
Uber	Drivers	Riders
Airbnb	Hosts	Guests
YouTube	Creators	Viewers
Wikipedia	Editors (0.02% of users)	Readers
Slack	Workspace admins/champions	Team members
Dating apps (Tinder)	Women (control matching)	Men
Marketplaces	Sellers/providers	Buyers
Workplace tools	Managers who author/invite	Team members
App stores	Developers	Users

The 1/10/100 Rule (from the book): For every 100 platform users, roughly 10 engage passively (comments, likes), and 1 creates content. The 1% who create are the hard side. This ratio varies by product but the concentration of value creation in a small minority is nearly universal in networked products. (medium.com/twosapp)

The hard side can shift over time. In early marketplaces, supply is the hard side. Once supply is abundant, demand becomes the constraint. In early creator platforms, creators are the hard side. Once the platform is credible, creators may start seeking it out, shifting the difficulty toward audience acquisition.

Uber's "Power Driver" example: In any given market, Power Drivers constitute 20% of the driver supply but create 60% of all trips. These are among the most valuable users on the platform. Early Uber paid minimum hourly guarantees, referral bonuses, and other subsidies specifically to recruit and retain this 20%. (andrewchen.com/solve-a-hard-problem-cold-start-problem/)

1.4 The Five Stages of Cold Start Theory

Chen's "Cold Start Theory" maps the full lifecycle of a networked product as an S-curve with five distinct phases. The key warning: **strategies that work at Stage 1 fail at Stage 3.** The playbook must evolve.

Stage 1: The Cold Start Problem The founding challenge. Network effects are actively working against you — the product is empty, and empty products create terrible experiences that drive churn. The task is to build the first atomic network: assemble the right users in the right configuration simultaneously. This is a manual, unscalable, founder-driven process. (whatyouwilllearn.com)

Stage 2: The Tipping Point The first atomic network is stable, but you need to replicate it across new markets. The product begins to tip — "city by city, company by company, campus by campus." The goal is to find repeatable playbooks for building more atomic networks, not just maintaining the first one. Each new network tips faster than the last as the playbook is refined and the product's reputation spreads. (medium.com/twosapp)

Tipping point tactics that worked historically:

- **Invite-only strategy** (LinkedIn, Gmail, Facebook): Exclusivity creates density among well-connected early users. The reason this works is not just hype — invite mechanics cause networks to self-replicate. Each invited user is already connected to their inviter and likely others they know. (sachinrekhi.com)

- **"Come for the tool, stay for the network"** (Instagram, Dropbox, LinkedIn): Attract users with standalone single-player utility, then gradually surface network features as the user base grows. Tools drive acquisition; networks drive retention.
- **Geographic/community focus** (Yelp in SF, Yammer in single companies, Twitter at SXSW, Snapchat in Southern California): Start where existing density already exists, rather than launching nationally.

Stage 3: Escape Velocity Network effects activate and begin compounding. Three "flywheels" strengthen simultaneously:

- **Acquisition Effect:** Existing users recruit new users, reducing CAC over time
- **Engagement Effect:** A denser network means more content, connections, and interaction — the product gets stickier
- **Economic Effect:** Improved unit economics — better matching, lower operational costs, higher conversion rates, premium pricing power

This is when paid growth begins to scale, referral programs become self-sustaining, and network density generates its own momentum. (sachinrekhi.com)

Stage 4: Hitting the Ceiling Every networked product eventually hits natural growth limits. Causes include:

- Market saturation (you've acquired most eligible users in a segment)
- Rising CAC as easiest users are already acquired
- Quality degradation — fraud, spam, trolls, low-quality content as the network opens up
- Multi-tenanting — users using multiple competing platforms simultaneously
- Commoditization — competitors achieve sufficient network density in the same market

Each ceiling requires a different response. Market saturation calls for new segment expansion. Quality degradation calls for trust and safety investment. CAC increases call for new channels.

Stage 5: The Moat At scale, network effects create genuine competitive defensibility. Competitors face the same cold start problem you already solved — they must convince users to abandon an established network to join a sparse new one. But Chen emphasizes: **the moat must be actively defended.** It degrades without continued investment in hard-side quality, product improvement, and new network formation.

Chen's case study: Google+ failed to displace Facebook despite 90 million users because the networks were sparse and low-engagement — users had all their real connections on Facebook, not Google+. Network size is meaningless without network density. (medium.com/twosapp)

Part 2: A Practical Taxonomy of Cold Start Types

Chen's framework applies broadly to all networked products, but the specific cold start challenge varies significantly by product type. Synthesizing Chen's work with practitioner frameworks, there are five functionally distinct cold start types. A given product may face one or several simultaneously.

Type 1: The Network Cold Start (Chicken-and-Egg for Two-Sided Networks)

What it is: The classic chicken-and-egg problem. You have two distinct user groups that each need the other to be present before they see value. Neither side has incentive to join first without the other. This is the defining cold start problem for two-sided platforms.

Products that face this: Marketplaces (buyers/sellers), ride-sharing (riders/drivers), home-sharing (guests/hosts), dating apps (women/men), app stores (users/developers), staffing platforms (employers/candidates).

Why it's hard: NFX puts it precisely: "To break them apart you must have a better value proposition for both parties simultaneously, or else nobody moves." (nfx.com/post/network-effects-bible)

Diagnosis questions:

- Does your product have two meaningfully distinct user groups?
- Does each group need the other to be present to get value?
- Would one group sign up if the other wasn't there yet?
- Which group has more to lose by joining an empty network?

Real examples:

- **Airbnb:** Guests won't book if there are no listings. Hosts won't list if there are no guests. Airbnb solved this by launching at SXSW (a dense event with a temporary housing shortage) and later by having early employees photograph listings in NYC to improve supply quality and attract more hosts. Supply came first, always.
- **OpenTable:** Restaurant reservations required both restaurant software and diner adoption. OpenTable solved the restaurant side first (installing free hardware), then built diner demand once restaurant coverage was sufficient. The supply side required multi-year investment before demand acquisition made sense.
- **Uber:** Drivers don't drive if there are no riders; riders don't return if there are no drivers. Uber subsidized drivers with minimum hourly guarantees and recruited the first riders through events and free rides, then built out demand once supply density was achieved in a market.

Core strategy: Decide which side is your hard side. Concentrate nearly all early resources on solving the hard side's specific problems first. Use subsidies, direct recruitment, and manual operations to achieve hard-side density before spending on demand. The order of operations for most marketplaces: supply → demand → supply → supply → supply. (lennysnewsletter.com/p/how-to-kickstart-and-scale-a-marketplace-2e5)

Type 2: The Content/Supply Cold Start (Empty Platform)

What it is: The platform is technically functional but has no content, no supply, and no inventory. Users arrive, find nothing, and leave. The issue isn't that two sides need each other — it's that the platform's value proposition is entirely dependent on content that doesn't exist yet.

Products that face this: Content platforms (YouTube, TikTok, Reddit), review platforms (Yelp, Glassdoor), knowledge bases (Wikipedia, Stack Overflow), job boards, listing sites, Q&A platforms.

Why it's different from the network cold start: In a content cold start, you don't necessarily need a "demand side" to create supply — creators and contributors can post into the void with some motivation. The challenge is bootstrapping that creator behavior before there's an audience to reward it.

Diagnosis questions:

- Is your platform's value derived from user-generated content or listings?
- Would users see value on day one if the content was already there?
- Is the content creation side (supply) decoupled from the consumption side (demand)?
- Can you create or seed the content yourself in early stages?

Real examples:

- **Reddit:** Early Reddit founders (Steve Huffman and Alexis Ohanian) created hundreds of fake accounts and posted content themselves to make the platform look active before real users arrived. They seeded the illusion of a community to get a real community started.
- **Yelp:** Yelp launched exclusively in San Francisco, saturating one city's restaurant reviews before expanding. They recruited their first reviewers from their own network — people with opinions who would write detailed reviews before a reader audience existed.
- **YouTube (early):** YouTube bootstrapped content by embedding viral videos that existed elsewhere on the internet (including copyrighted content — which created its own problems). The first "real" YouTube community was skateboarding, snowboarding, and extreme sports, where creators already had content and motivation.
- **GrubHub:** GrubHub solved an empty platform problem by manually scanning and uploading restaurant menus — supply the founders themselves created — which drove SEO traffic and organic demand. (lennysnewsletter.com/p/how-to-

kickstart-and-scale-a-marketplace-2e5)

- **Stack Overflow:** Launched with a seeded set of programming questions and answers by the founders, building enough initial content to attract organic search traffic and contribute-motivated early users.

Core strategies:

1. **Wizard of Oz / Fake it till you make it:** Founders manually create content, fake activity, or manually fulfill requests while the automated systems are built out. Customers don't know the process is manual.
2. **Recruit hard-side creators directly:** Find the people who already have content or expertise and recruit them individually to your platform before you have an audience for them. Promise future audience.
3. **Seed with your own content:** Post it yourself. Hire contractors. Import public domain content. Anything to populate the empty state.
4. **Create single-player value for creators:** Give creators tools they can use even without an audience — analytics, editing tools, hosting — so they have a reason to be on the platform before their audience arrives.

Type 3: The Marketplace Cold Start (No Buyers + No Sellers)

What it is: A specific and very common variant of the two-sided network cold start. Marketplaces require simultaneous inventory/supply (sellers, service providers, hosts) and shoppers/buyers to create any transaction. Neither side has reason to invest without the other. The cold start isn't just an empty platform — it's an empty market where no economic activity can occur.

Note: Some practitioners treat marketplace cold start as a subset of network cold start (Type 1). The distinction is worth making separately because marketplaces have specific dynamics: transactions are the value unit (not connections), supply is typically the critical constraint, and "multi-tenanting" (sellers using multiple platforms) is common and affects strategy.

Products that face this: Physical goods marketplaces (Etsy, eBay, Amazon), service marketplaces (TaskRabbit, Thumbtack, Rover, Upwork), rental marketplaces (Airbnb, Turo, VRBO), delivery marketplaces (DoorDash, Instacart).

Diagnosis questions:

- Is the core value unit a transaction between buyers and sellers, not just a connection?
- Can sellers independently market their listings to drive their own buyers?
- Is the supply constraint geographic or category-specific?
- How often do sellers use multiple competing platforms simultaneously?

Real examples with specific tactics:

- **Etsy:** Founders showed the prototype at physical craft fairs, recruiting sellers one by one. Sellers are inherently motivated to find buyers for their work and did grassroots marketing to their existing communities. Supply drove demand. (lennysnewsletter.com/p/how-to-kickstart-and-scale-a-marketplace-2e5)
- **DoorDash:** Founders created their own Dasher operation with themselves as the first dashers, running deliveries manually to validate demand before building the platform. "We did everything that wasn't going to scale." Restaurants printed stickers and placards without DoorDash prompting them — supply marketing for demand.
- **TaskRabbit:** Used direct outreach and street teams to recruit "Rabbits" (taskers). First customers were acquired entirely through PR and WOM. Taskers found their first clients through TaskRabbit-organized events. Founded partnership with Gap, Sephora, and Pepsi to generate initial task demand.
- **Thumbtack:** 80–90% of early growth came from SEO — creating pages for every service category in every city, generating Google traffic before the marketplace existed, then backfilling supply. (lennysnewsletter.com/p/how-to-kickstart-and-scale-a-marketplace-2e5)

The key tactical insight from Lenny Rachitsky's research: Most successful marketplaces prioritized supply first, not demand. Supply can be recruited through direct sales. Demand follows supply naturally if the price/quality proposition is right. The exception is when demand is the rarer commodity (e.g., enterprise buyers who are hard to reach).

Supply-drives-demand examples:

- **Patreon:** Creators marketed their own pages to their audiences. Patreon got creators; creators brought the patrons.
- **DoorDash restaurants:** Restaurants put DoorDash stickers in their windows, driving organic consumer discovery.
- **Lyft:** Drivers wore pink mustaches, generating consumer WOM about a novel product.

Type 4: The Group/Invite Cold Start (Quorum Required)

What it is: The product is only useful when a specific group of people — often a pre-existing social or professional group — all adopt it together. The value isn't just that there are many users; it's that *your people* are on it. You need a quorum within a specific social context. Individual adoption is worthless.

Products that face this: Team communication tools (Slack, Microsoft Teams), group coordination apps (Doodle, Calendly), multiplayer games, family apps, friend group apps (Houseparty, BeReal), school coordination tools, workplace tools that require team-level adoption.

Why it's distinct: Unlike a marketplace or content platform, there's no "supply side" to seed separately. The entire value of the product collapses if your specific group doesn't adopt it. And users can't be randomly aggregated — it has to be *this* team, *this* friend group, *this* class, *these* coworkers.

Diagnosis questions:

- Does your product require a pre-existing relationship between users (not just any users)?
- Would a single user on your platform see zero or near-zero value?
- Does value require a specific minimum group size (quorum)?
- Is the relevant unit a team, household, friend group, or other pre-existing social unit?
- Would users be willing to recruit their specific group in order to use the product?

Real examples:

- **Slack:** Slack's atomic network was a single company team. One person adopting Slack has no value. One team adopting Slack in a single company has full value — they can communicate. Slack grew by getting one team at one company to adopt, then spreading to adjacent teams and companies through employee movement. (lennysnewsletter.com/p/atomic-network)
- **WhatsApp:** Needed groups of friends and family to all be on the same platform. Solved this in developing markets where SMS was expensive — the cost savings were sufficient motivation for people to recruit their entire networks.
- **Houseparty:** Required groups of friends to all join simultaneously for video hangouts. Solved the group cold start by having groups sign up together, creating shared accounts and notifications that pulled the whole group in. Failed post-COVID when the group video habit faded. (startupik.com)
- **Clubhouse:** Required both speakers and listeners in the same "room." Solved with invite-only mechanics (creating group-level FOMO) and early celebrity speaker recruitment (Elon Musk, Mark Zuckerberg). The velvet rope strategy: when your invitee joins, they immediately get an invite to give out — mechanic that copies the social graph one link at a time. (growthmodels.co)
- **Yammer:** Enterprise communication that started within single companies. Solved by targeting companies where the founders already had connections, getting one team to adopt inside a company, then spreading virally inside the organization.

Core strategy: The invite mechanic is specifically designed for this cold start type. As Chen explains: invite mechanics cause networks to self-replicate — each new user is already connected to their inviter, so the graph copies itself organically. Invite-only also creates density before opening broadly, ensuring the first users don't arrive to an empty network.

Minimum viable group size matters: For Zoom, 2 people. For Slack, 3+ people. For group chat apps, 5+ people. For community apps, potentially much larger. The key design question is: what's the smallest group that experiences full value?

Type 5: The AI / Algorithm Cold Start (Training Data Problem)

What it is: The product's core value proposition depends on personalization, recommendation, or intelligence that requires data — but you have no data when you launch. The algorithm is cold, not just the network. This is increasingly relevant as AI-powered products become the default.

Products that face this: Recommendation engines (Spotify, Netflix, TikTok), personalized feeds (news apps, social feeds), matching algorithms (Hinge, LinkedIn jobs), AI assistants trained on user behavior, fraud detection systems, content moderation systems.

Why it's distinct from other cold starts: Even if you have users, the product fails to deliver personalized value until you have *enough data about each individual user* and *enough aggregate data to train your models*. A new user gets generic, non-personalized recommendations, which feels worse than Google or a manual search. The cold start is both systemic (not enough aggregate data) and per-user (new user has no history). (wikipedia.org)

Three sub-types (from recommender systems literature):

1. **New user cold start:** A new user registers and the system has no preference data. Spotify asks you to pick 3 favorite artists on signup. Pandora has you rate songs. Netflix asks you to select genres.
2. **New item cold start:** New content, listings, or items are added that have zero interaction history. Collaborative filtering can't recommend them without ratings.
3. **System cold start:** The entire system is new and has no interaction data at all. (This is the founding cold start.)

Diagnosis questions:

- Does your product's value degrade without user data/history?
- Would a new user experience a meaningfully worse product than a veteran user?
- Does your product require aggregate data across many users to work (collaborative filtering)?
- What's the minimum data density required for the algorithm to function acceptably?

Solutions used by real products:

- **Spotify:** Onboarding questionnaire asking for favorite artists and genres — explicit preference data to bootstrap the cold user
- **Netflix:** Genre-based onboarding, followed by explicit rating prompts
- **TikTok:** Showed all new users the same viral content to generate interaction data fast, then personalized rapidly. Their cold start experience is famously short — the algorithm needs only ~5–10 interactions to start personalizing.
- **Hinge:** Uses detailed profile data (prompts, preferences, dealbreakers) as a proxy for matching signals before interaction history exists
- **YouTube early algorithm:** Used view count and recency as crude proxies before engagement signals were available

For AI-powered student products: The cold start problem means your AI features may work poorly for early users. This is both a technical problem (insufficient training data) and a UX problem (users don't understand why the AI seems dumb). The solution is usually to be explicit about the ramp ("the more you use it, the smarter it gets") or to pre-populate with synthetic data, curator inputs, or defaults that feel acceptable before personalization kicks in.

Part 3: The NFX Framework — 16 Network Effect Types

NFX (the network effects-focused VC firm) offers a complementary taxonomy organized around the *type of network effect* rather than the type of cold start. Their research identified 16 distinct network effect types, organized into 5 categories by strength. Understanding which type your product has helps determine how severe the cold start problem is and how defensible the business becomes once you solve it. (nfx.com/post/network-effects-manual)

NFX's 5 Categories of Network Effects (Strongest to Weakest)

Category 1: Direct Network Effects Value increases directly as more users of the same type join.

- **Physical:** Infrastructure-based (roads, broadband, landlines). Cold start: extremely capital-intensive, often requires regulation or subsidy.
- **Protocol:** Shared standard (Bitcoin, TCP/IP, Ethernet). Cold start: requires developer/adopter coordination around a standard.
- **Personal Utility:** Essential daily communication tied to identity (iMessage, WhatsApp). Cold start: requires recruiting specific people in your life.
- **Personal:** Identity and reputation networks (Facebook, Twitter). Cold start: requires reaching social graph density.
- **Market Networks:** Many-to-many with transaction focus (HoneyBook, AngelList). Cold start: requires recruiting professionals in a specific category.
- **Hub-and-Spoke:** Content submitted to a central hub that distributes selectively (TikTok algorithm, YouTube). Cold start: requires creators who will post even with few viewers.

Category 2: Two-Sided Network Effects Two distinct user classes create complementary value.

- **Marketplace (2-Sided):** Buyers and sellers (eBay, Craigslist). Cold start: classic chicken-and-egg, simultaneous supply and demand needed.
- **Platform (2-Sided):** Developers build for the platform (iOS, Xbox). Cold start: requires developer ecosystem plus user base simultaneously.
- **Asymptotic Marketplace:** Value gains diminish after critical threshold (Uber, Lyft). Cold start: once basic coverage exists (~4 min wait time), additional supply adds minimal value. Paradoxically easier to compete against incumbents.

Category 3: Data Network Effects Product improves with accumulated user data.

- **Data NfX:** Usage generates useful new data continuously (Waze, Yelp). Cold start: product is worse than alternatives until data accumulates.

Category 4: Tech Performance Network Effects Product performance improves as more users join.

- **Tech Performance:** Speed/cost improves with network size (BitTorrent). Cold start: product may be slow/expensive at launch but improves with scale.

Category 5: Social Network Effects (Weakest, but fastest to build) Psychology, language, and social belonging drive adoption.

- **Language:** Shared vocabulary becomes more valuable with adoption (Google as verb, Xerox as verb).
- **Belief:** Self-reinforcing belief creates value (Bitcoin, religions). Cold start: requires seeding belief in a founding community.
- **Bandwagon:** Fear of missing out drives adoption (Slack in tech, Apple). Cold start: create cultural momentum and buzz.
- **Tribal:** Identity-based in-group value (alumni networks, military units, Y Combinator). Cold start: requires founding the tribe and defining who's in.

NFX's key diagnostic question for cold start severity: How much simultaneous density do you need before the product works at all? Physical infrastructure needs millions. WhatsApp needs a handful. Bandwagon effects just need a cool founding community. The harder the cold start, the more defensible the moat — but the more capital-intensive the launch. (nfx.com/post/network-effects-bible)

Part 4: Diagnostic Framework — Identifying Your Cold Start Type

Synthesizing Andrew Chen, NFX, and Lenny Rachitsky, here is a practical diagnostic framework for student teams.

Diagnostic Question Tree

Step 1: Does your product have network effects at all?

- Does your product become more valuable as more people use it?
- If NO: You don't have a cold start problem — you have a product distribution problem. Different playbook.

- If YES: Continue to Step 2.

Step 2: What kind of network does your product create?

Ask: "Who are the distinct user types on my platform?"

- **One user type (everyone is the same):** You have a direct network (social, communication, community). → Cold start Type 1 (network) or Type 4 (group/invite)
- **Two user types (supply and demand):** You have a two-sided market. → Cold start Type 1 (marketplace/chicken-and-egg) or Type 3 (marketplace)
- **No user-user connection, but data-driven personalization:** You have data/algorithm network effects. → Cold start Type 5 (AI/algorithm)

Step 3 (for two-sided networks): Who needs whom first?

- Which side is your hard side? (Who creates the value? Who has more options elsewhere?)
- Would you launch with 1,000 of the supply side and 0 of the demand side, or vice versa?
- The hard side gets your first 90% of effort.

Step 4 (for one-sided networks): What is the minimum viable group size?

- Can one person get value? → It's a tool, not a pure network product. Use "come for the tool" strategy.
- Does it require a specific pre-existing group (team, friends, family)? → Type 4 (group/invite)
- Can value come from any strangers? → Type 1 (network) or Type 2 (content)

Step 5: Is the platform empty of content?

- Is the value of your product content that users create?
- Would your product be valuable if you filled it with professional content?
- If yes to either: You have a content cold start dimension (Type 2) regardless of whether you also have a network cold start.

Step 6: Does your product use AI/personalization?

- Does the user experience degrade significantly for new users who have no history?
- Is there a recommender, matching algorithm, or personalization engine that needs data to work?
- If yes: You have an AI cold start dimension (Type 5) on top of any network cold start.

Diagnostic Summary Table

Product Type	Cold Start Type	Hard Side	Primary First Move
Marketplace (physical goods)	Two-sided (Type 3)	Sellers	Recruit supply manually
Marketplace (services)	Two-sided (Type 3)	Service providers	Recruit supply directly
Ride/delivery	Two-sided (Type 1)	Drivers/dashers	Subsidize supply, guarantee income
Social network	Network (Type 1)	Power users/creators	Invite-only or geo-focused launch
Content platform	Content (Type 2)	Creators	Seed content, recruit creators
Team/workplace tool	Group/invite (Type 4)	Admins/champions	Get one whole team, then expand
Group communication	Group/invite (Type 4)	Group organizer	Recruit the whole group at once

Dating app	Two-sided (Type 1)	Women (typically)	Seed women first, address trust/safety
Review/rating site	Content (Type 2)	Reviewers/editors	Seed reviews, recruit contributors
AI-powered tool	AI/algorithm (Type 5)	Data generators	Onboarding questionnaire, synthetic data
Community/forum	Content + Network	Power posters	Seed with curated founding community

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Part 5: Product-Specific Strategies by Cold Start Type

Strategy 1: Constrained Launch (Geographic/Community Focus)

Works for: Network cold start, Marketplace cold start Examples: Uber (SF only), Facebook (Harvard), Yelp (San Francisco), Yammer (single companies), Twitter (SXSW), Snapchat (Southern California high schools)

Principle: Network density beats network size. A product that works perfectly for 1,000 people in one community will spread from that community better than a product that works barely at all for 1,000,000 people spread across the country.

Chen: "You'd much rather have 100,000 users with a density of 30 connections/person than 1,000,000 who have a density of 2 connections." (andrewchen.com/how-to-solve-the-cold-start-problem-for-social-products/)

For student teams: Define your atomic network before you define your TAM. What is the smallest group of people who would experience full value from your product if they all adopted it together?

Strategy 2: Single-Player Mode / "Come for the Tool"

Works for: Content cold start, Group cold start, Network cold start Examples: Instagram (filters → social), LinkedIn (resume hosting → professional network), Pinterest (visual bookmarking → social boards), Dropbox (file sync → sharing), GitHub (version control → collaboration), Glassdoor (job listings → reviews)

Principle: Give users value they can extract alone, before the network exists. Then gradually surface network features as population density increases.

The move from tool to network is not automatic — it requires intentional design. Instagram originally had no "follow" feature and no feed. These were added after the tool-driven user base was established. (andrewchen.com/how-to-solve-the-cold-start-problem-for-social-products/)

For student teams: Ask whether your product has any standalone utility for a single user. If not, build some. The standalone utility doesn't have to be your core value proposition — it just has to be enough to get users into the product and make the first session non-empty.

Strategy 3: Supply Seeding / Wizard of Oz

Works for: Marketplace cold start, Content cold start Examples: Airbnb (craigslist cross-posting and professional photography), Reddit (founders created fake accounts), GrubHub (manually scanned menus), Amazon (Bezos personally found and listed first books)

Principle: Manually create, seed, or bootstrap the supply side before the platform can generate it on its own. The supply doesn't have to be real-network supply — it just has to look and feel complete enough that demand side users get value.

For student teams: Before building automated supply mechanics, ask whether a founder can manually fill the supply role. If you're building a marketplace for freelance designers, can a founder recruit 10 designers personally? If you're building an events discovery platform, can you add events yourself for the first month? Manual supply is not a bug — it's a valid founding strategy.

Strategy 4: Invite Mechanics / Velvet Rope

Works for: Group cold start, Network cold start Examples: LinkedIn, Gmail, Clubhouse, Superhuman, Discord servers, Notion (invite-only for teams)

Principle: Invite-only creates three advantages simultaneously: (1) early network density among well-connected users, (2) better welcome experience because invitees already have at least one connection, (3) FOMO and perceived exclusivity that makes the uninvited want in.

The mechanic: Each invited user gets invites to give out, causing the network to copy itself one social link at a time. This is not just a growth hack — it's a structural solution to the group cold start problem. You're not growing randomly; you're growing along the edges of existing social graphs. (sachinrekhi.com)

For student teams: Consider whether limiting your first wave of users to a specific invitation-only cohort (e.g., "apply to join the beta") would result in a more engaged founding community. Even 50 highly engaged users in an invite-only beta outperforms 500 users who signed up and never returned.

Strategy 5: Piggyback on an Existing Network

Works for: Network cold start, Content cold start Examples: Instagram → Facebook (early photos cross-posted automatically), PayPal → eBay (PayPal payment button on every eBay listing), Airbnb → Craigslist (auto-posting listings to drive traffic back to Airbnb), Zynga → Facebook (game notifications in the news feed)

Principle: Find a network where your target users already exist and give them a reason to pull those connections into your product. You're not building a network from scratch — you're siphoning an existing one.

Risks: Platform dependency. If the host platform changes its APIs or kicks you off, your growth channel disappears. PayPal was banned by eBay, then acquired by eBay. Airbnb's Craigslist scraping was never officially sanctioned. Instagram's Facebook integration was blessed by Facebook (which later acquired Instagram). Know your relationship with the host platform. (medium.com/samaipata-ventures)

Part 6: Warning Signs and Death Spirals

Early Warning Signs That Cold Start Strategy Isn't Working

1. **Low D1 retention** — New users aren't coming back the next day. The first-session experience wasn't good enough to pull them back. Root cause: the atomic network wasn't complete when they arrived.
2. **High activation rate, low engagement** — Users create accounts, poke around, and leave. They understand what the product does but don't find anything to do. Root cause: content or supply cold start — the platform is technically functional but empty.
3. **High churn among early cohorts** — The users who signed up in the first weeks are leaving faster than new users are arriving. Root cause: the founding community didn't have enough connections to stay engaged.
4. **Network clusters but doesn't spread** — You have one dense cluster of users (your friends, a single company, one campus) and the product doesn't expand beyond that cluster. Root cause: atomic network was built successfully but tipping point strategy hasn't been developed.
5. **No organic referrals** — Your users like the product but don't bring others. Root cause: the product isn't creating the network effect that would motivate referrals. May indicate missing viral loop design.
6. **Supply side attrition** — Your marketplace providers, creators, or contributors are signing up but then going inactive. Root cause: insufficient demand for their supply. The easy side hasn't arrived in sufficient numbers to reward the hard side's participation.

The Death Spiral

The cold start death spiral: users arrive → network is too sparse to deliver value → users leave → network gets sparser → worse experience for remaining users → they leave too → platform fails.

Chen: "At their inception, network effects are actually a destructive force where new users churn because not enough other users are there yet." (sachinrekhi.com)

Escape Velocity Signals

You've broken through the cold start when:

- D1/D7/D30 retention is climbing, not flat or declining
- Users are generating organic referrals without incentive
- The hard side is staying active without subsidies
- Word-of-mouth is a measurable acquisition channel
- The product is growing in one atomic network while you're launching a second
- CAC is declining as organic acquisition increases

For Chen's benchmark: Facebook considered a user "activated" when they had 10 friends within 7 days. That 10/7 metric became their proxy for whether the cold start was solved. (medium.com/twosapp)

Part 7: Application to Student Teams — Diagnostic Playbook

For Student Teams with ~14 Days to Feature Freeze

The goal is not to solve the cold start problem in 14 days. The goal is to:

1. Correctly diagnose which type of cold start you face
2. Execute the one or two tactics that are realistic given your resources
3. Validate that your chosen strategy is working or pivot before demo day

Quick Diagnostic for Each Student Product Type

If you're building a marketplace:

- Who is your hard side? (usually supply/providers)
- Have you manually recruited at least 5–10 hard-side users?
- Can those users generate value for demand-side users today?
- If no: spend 100% of your time recruiting supply manually

If you're building a social/community product:

- What is your atomic network? (Who are the 20–50 people who would create a self-sustaining micro-community?)
- Are those people connected to each other already, or just connected to you?
- Have you recruited enough of them to make the first sessions non-empty?
- If no: run an invite-only closed beta with a specific founding community

If you're building a group/team tool:

- Have you gotten at least one complete team (5–10 people with pre-existing relationships) to adopt it?
- Are all of them using it, or just the one advocate?
- If not: your activation target is getting one whole team through onboarding, not just individuals

If you're building a content platform:

- Is there any content there for a new user to engage with?

- If not: can you seed it yourself? Can you recruit 3–5 early contributors?
- What's the minimum content density that makes a first session feel not-empty?

If you're building an AI-powered tool:

- What does a brand-new user (no history, no data) experience?
- Is that experience good enough to bring them back?
- If not: what's your cold start content or onboarding questionnaire that fills in the gap?

The Only Test That Matters at This Stage

Regardless of cold start type, the one test: **Give your product to someone who doesn't know you and has no obligation to be nice. Watch what they do. Do they come back the next day?**

If yes: your cold start solution is working. If no: diagnose why the first session failed to deliver value and trace it back to which cold start type is the root cause.

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